cp_8a

1

Print Length times (For Loop) | 30/30

Solved

Easy

Write a program that reads a string and prints the first character (The character at index 0) of the given string on `N` lines, where `N` is the length of the given string.

Note

Use the **For** loop to iterate over the string.

Input

The input will be a single line containing a string.

Output

The output should be `N` lines containing a string that is the first character of the given string, each on a new line.

Explanation

For example, if the given string is **query**,

- The first character in the string **query** is **q**.
- The length of the string **query** is **5**.
- The character **q** should be printed on **5** lines.

The output should be,

```
q
q
q
q
q
```

Sample Input 1

```
query
```

Sample Output 1

```
q
q
q
q
q
```

code:

```
name=input()
for i in range(len(name)):
```

```
    print(name[0])
```

2

Print the Characters (For Loop) | 35/35

Easy

Write a program that reads a string and prints each character of the given string on a new line.

Note

Use the **For** loop to iterate over the string.

Input

The input will be a single line containing a string.

Output

The output should contain all the characters of the given string with each character on a new line.

Explanation

For example, if the given string is **python**,

- The characters in the string **python** are **p**, **y**, **t**, **h**, **o**, and **n**.

The output should be,

```
p
y
t
h
o
n
```

Sample Input 1

```
python
```

Sample Output 1

```
p
y
t
h
o
n
```

code:

```
name=input()
for alphabet in name:
```

```
    print(alphabet)
```

3

Print N Integers (For Loop) | 30/30

Write a program that reads a number `N` and prints numbers from **1** to `N` .

Note

Use the **For** loop to iterate over the range of numbers.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should contain all numbers from **1** to `N` , each on a new line.

Explanation

For example, if the given number is `N = 3` ,

- The numbers from **1** to **3** are **1**, **2**, and **3**.

The output should be,

```
1
2
3
```

Sample Input 1

```
3
```

Sample Output 1

```
1
2
3
```

Sample Input 2

```
5
```

Sample Output 2

```
1
2
3
4
5
```

code:

```
n=int(input())
for i in range(1,n+1):
```

```
    print(i)
```

```
4
```

First N Natural Numbers (For Loop) | 35/35

Easy

Write a program that reads a number `N` and prints Natural Numbers from 1 to `N` .

Note

- All the positive numbers from 1 are called **Natural Numbers**.
- Use the **For** loop to iterate over the range of numbers.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should contain all the Natural Numbers from 1 to `N` , each on a new line.

Explanation

For example, if the given number is `N = 7` ,

- The Natural Numbers from 1 to 7 are 1, 2, 3, 4, 5, 6 and 7.

The output should be,

```
1
2
3
4
5
6
7
```

Sample Input 1

```
7
```

Sample Output 1

```
1
2
3
4
5
6
7
```

code:

```
num=int(input())
for i in range(1,num+1):
    print(i)
```

5

Sum of N Natural Numbers (For Loop) | 30/30

Write a program that reads a number `N` and prints the sum of Natural Numbers from **1** to `N` .

Note

Use the **For** loop to iterate over the range of numbers.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the sum of Natural Numbers from **1** to `N` .

Explanation

For example, if the given number is `N = 6` ,

- The Natural Numbers from **1** to **6** are **1**, **2**, **3**, **4**, **5**, and **6**.
- The sum of **6** Natural Numbers is **21**. (`1 + 2 + 3 + 4 + 5 + 6 = 21`)

Number	Previous Sum	Previous Sum + Number = Current Sum
1	0	0 + 1 = 1
2	1	1 + 2 = 3
3	3	3 + 3 = 6
4	6	6 + 4 = 10
5	10	10 + 5 = 15
6	15	15 + 6 = 21

The output should be **21**.

Sample Input 1

```
6
```

Sample Output 1

```
21
```

code:

```
num=int(input())
sum=0
```

```
for i in range(num+1):
    sum=sum+i
print(sum)
```

```
6
```

Average of the N numbers (For Loop) | 30/30

Easy

Write a program that reads a number `N` and prints the average of numbers from **1** to `N` .

Note

- The average of numbers from **1** to `N` can be calculated as,
  Average = Sum of numbers from **1** to `N` / Count of numbers (`N`)
  Example: If `N = 3` , the average of numbers from **1** to **3** is,
  Average = (1 + 2 + 3) / 3 = 2.0
- Use the **For** loop to iterate over the range of numbers.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing a float that is the average of numbers from **1** to `N` .

Explanation

For example, if the given number is `N = 8` ,

- The numbers from **1** to **8** are **1**, **2**, **3**, **4**, **5**, **6**, **7**, **8**.
- The count of numbers from **1** to **8** is **8**.
- The sum of the numbers is **36**. (`1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36`)
- The average of the numbers is **4.5**. (`36 / 8 = 4.5`)

The output should be **4.5**.

Sample Input 1

```
8
```

Sample Output 1

```
4.5
```

code:

```
num=int(input())
sum=0
```

```
for i in range (1,num+1):
    sum=sum+i
avg=sum/num
print(avg)
```

```
7
```

Cubes of N Numbers (For Loop) | 30/30

Easy

Write a program that reads a number `N` and prints the cube of numbers from 1 to `N` .

Note

- The cube of a number `X` is `X` power 3.(X^3)
- Use the **For** loop to iterate over the range of numbers.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` lines containing integers that are the cubes of numbers from 1 to `N` .

Explanation

For example, if the given number is `N = 3` ,

- The numbers from 1 to 3 are 1, 2 and 3.
- The cubes of numbers from 1 to 3 are 1, 8, and 27.

Number (N)	Cube of N (N^3)
1	1
2	8
3	27

The output should be,

```
1
8
27
```

Sample Input 1

```
3
```

Sample Output 1

```
1
8
27
```

code:

```
num=int(input())
for i in range(1,num+1):
    cube=i**3
    print(cube)
```

8

Read N Inputs (For Loop) | 30/30 ● Solved

Given an integer N, write a program which reads N inputs and prints them.
Note: Use For Loop

Input

The first line of input will contain a positive integer, N.
The following N lines will contain an integer in each line.

Output

The output should be N lines, containing an integer per line.

Explanation

For example, if the given N is 3, then read the inputs in the next 3 lines and print them. If the given input integers in the next three lines are 8, 11, and 25, the output should be

```
8
11
25
```

Sample Input 1

```
3
8
11
25
```

Sample Output 1

```
8
11
25
```

Sample Input 2

```
2
7
20
```

Sample Output 2

```
7
20
```

code:

```
num=int(input())
for i in range(num):
ip_num=int(input())
print(ip_num)
```

```
9
```

Product of N Inputs (For Loop) | 30/30

Medium

Given an integer `N` , write a program that reads `N` inputs and prints the product of the given inputs.

Note

Use the **For** loop to iterate over the range of numbers.

Input

The first line of input contains an integer representing `N` .
The next `N` lines of input contain integers.

Output

The output should be a single line containing an integer that is the product of the given inputs.

Explanation

For example, if the given number is `N = 3` ,

- The **3** inputs are **2**, **3**, and **7**.
- The product of the given inputs is **42**. (`2 * 3 * 7 = 42`)

Number	Previous Product	Previous Product * Number = Current Product
2	1	1 * 2 = 2
3	2	2 * 3 = 6
7	6	6 * 7 = 42

The output should be **42**.

Sample Input 1

```
3
2
3
7
```

Sample Output 1

```
42
```

code:

```
num=int(input())
product=1
```

```
for i in range(num):
    ip_nums=int(input())
    product=product*ip_nums
print(product)
```

10

Solid Rectangle | 30/30 ● Solved

Given two integers M and N, write a program to print a solid rectangle pattern of M rows and N columns using the asterisk character (*).
Note: Use For Loop

Input

The first line of input will contain a positive integer.
The second line of input will contain a positive integer.

Output

The output should be the M number of lines and N number of columns with the asterisk character (*).
Note: There is a space after each asterisk character(*).

Explanation

For example, if the given numbers of rows are 2 and the number of columns is 3, the output should be 2 lines and 3 columns with the asterisk character(*).

```
* * *
* * *
```

For example, if the given numbers of rows are 4 and the number of columns is 6, the output should be 4 lines and 6 columns with the asterisk character(*).

```
* * * * * *
* * * * * *
* * * * * *
* * * * * *
```

Sample Input 1

```
2
3
```

Sample Output 1

```
* * *
* * *
```

code:

```
m=int(input())
n=int(input())
for m in range(m):
    print("* "*n)
```

11

Solid Right Angled Triangle | 30/30

Solved

Given an integer number (N) as input. Write a program to print the right-angled triangular pattern of N lines using an asterisk(*) character.
Note: Use For Loop

Input

The input will be a single line containing a positive integer (N).

Output

The output should be N lines with an asterisk(*) character printing in a right-angled triangular pattern.
Note: There is a space after each asterisk(*) character.

Explanation

For example, if the given number is 4,
the output should be

```
*
* *
* * *
* * * *
```

For example, if the given number is 2,
the output should be

```
*
* *
```

Sample Input 1

```
4
```

Sample Output 1

```
*
* *
* * *
* * * *
```

code:

```
n=int(input())
for i in range(1,n+1):
    print("* "*i)
```

12

Sum of the Given Numbers | 30/30 ● Solved

Given an integer N, write a program which reads N inputs and prints the sum of the given input integers.
Note: Use For Loop

Input

The first line of input will contain a positive integer, N.
The next N lines will contain the integers, each in a line.

Output

The output should be the sum of the given input integers.

Explanation

For example, if the given N is 3, then read the inputs in the next 3 lines and print the sum of the three input integers. If the given input integers in the next three lines are 8, 11, and 25, the output should be 44.

Sample Input 1

```
3
8
11
25
```

Sample Output 1

```
44
```

Sample Input 2

```
2
7
20
```

Sample Output 2

```
27
```

code:

```python
n=int(input())
sum=0
for i in range(n):
nums=int(input())
sum=sum+nums
print(sum)
```

13

Identify the Mistake - Styled Word | 10/10

Solved

Write a program that prints the individual characters of the given word separated with hyphens ("-").

Input

The input will be a single line containing a word.

Output

The output should be a single line containing characters separated with hyphens ("-").

Explanation

For example, if the given word is "Python", your code should print the individual characters in it separated with hyphens ("-"). So the output should be "P-y-t-h-o-n".

Sample Input 1

```
Python
```

Sample Output 1

```
P-y-t-h-o-n
```

Sample Input 2

```
Loops
```

Sample Output 2

```
L-o-o-p-s
```

code:

```
a = input()
len_of_a = len(a)
b = ""
for i in range(len_of_a):
    if i==(len_of_a-1):
        b=b+a[i]
    else:
         b = b  + a[i]+"-"
print(b)
```